vi. Biotin (Vitamin B_7):

Biotin is a heterocyclic monocarboxylic acid. It is a Sulphur-containing water soluble vitamin.

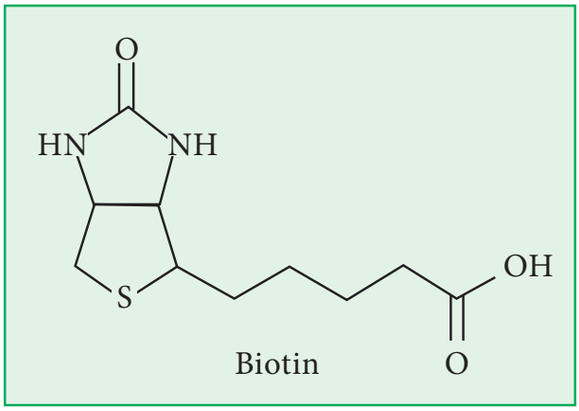

Figure 8.25 Structure of biotin

Sources:

Foods rich in biotin are liver, kidney milk and egg-yolk. Vegetables, grains are good sources.

Functions:

- It is essential for the synthesis of lipids.
- Biotin functions as the coenzyme for the enzyme called carboxylases, which catalyzes carboxylation reactions. Eg. Acetyl CoA carboxylase converts acetyl CoA to malonyl - CoA, which is required for fatty acid synthesis.

CO_2 + biotin- enzyme + ATP → Carboxy biotin - enzyme + ADP + Pi

Carboxy biotin enzyme + acetyl CoA $\xrightarrow{\text{Acetyl CoA carboxylase}}$ malonyl - CoA + biotin- enzyme

- It helps in the conversion of pyruvic acid to oxaloacetic acid. The enzyme that catalyzes this reaction is pyruvate carboxylase.
- It helps to maintain the skin and the nervous system in good condition.
- It involves in deamination of certain amino acids like aspartate, serine and threonine.

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